Bidirectional LSTM — Deep Recurrent Model

Per-model deep dive · learned temporal memory over the landmark sequence

Result: 0.433 accuracy · 0.429 weighted F1 · WER 0.567 · fold SD ±0.135
Notebook: 03_lstm_wlasl_landmark_baseline.ipynb

1 Theory

A Long Short-Term Memory network processes a sequence frame by frame, maintaining a gated memory cell that decides what to keep, forget and output. A **bidirectional** LSTM runs two of these — one forward, one backward — so the representation of any frame is informed by both its past and its future. This is the first model in the study to genuinely drop the HMM's frame-independence assumption: information flows directly from frame to frame through the memory, not through a single discrete stage.

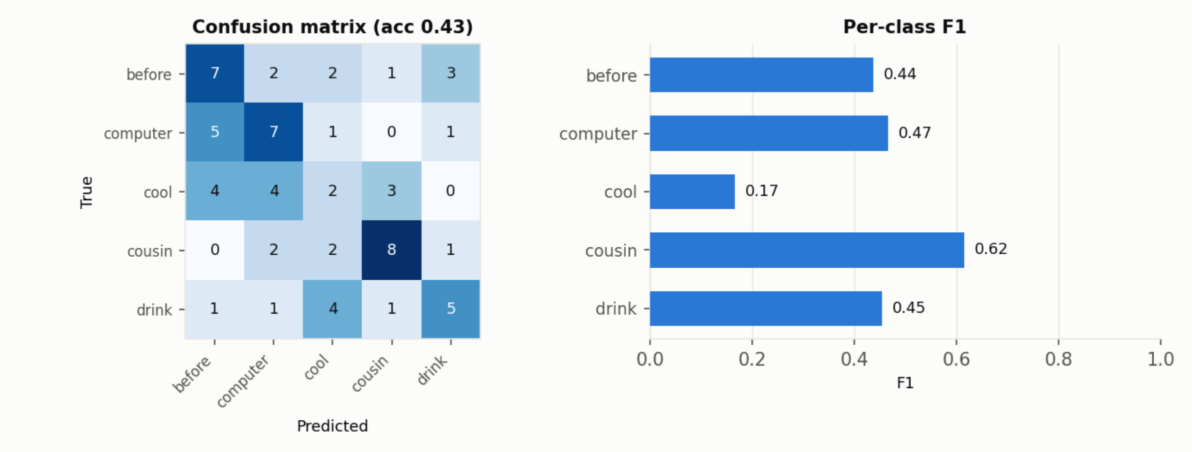
2 Method of execution

The 1,629-D landmark frames are passed through a linear input projection, then a 2-layer bidirectional LSTM (hidden size 128, dropout 0.3). The final forward and backward hidden states are concatenated and fed to a small MLP classifier. Padding frames are masked via packed sequences so the recurrence only sees real frames; training uses Adam with weight decay.

```
BiLSTM(input→proj→2×biLSTM(128)→concat(final states)→MLP→5 classes)
Adam(lr=1e-3, weight_decay=1e-4), dropout 0.3, masked/packed sequences
```

3 Results

Metric	Value (5-fold CV)
Accuracy	0.433
Weighted precision	0.428
Weighted recall	0.433
Weighted F1	0.429
Word Error Rate (WER)	0.567
Fold accuracies	0.43, 0.36, 0.69, 0.31, 0.38
Cross-val SD	±0.135
Distinct classes predicted	5 of 5
Majority baseline	0.224



BiLSTM confusion matrix and per-class F1 under 5-fold CV.

4 Why these results

At 0.433 the BiLSTM clears the baseline but posts **the highest fold-to-fold variance of any model** (SD ±0.135; folds: 0.43, 0.36, 0.69, 0.31, 0.38). Recurrent networks have many parameters and need substantial data to constrain them; with ~54 training videos the model lands in very different places depending on the split — one fold reaches 0.69, another 0.31. Its central tendency is modest not because recurrence is wrong for sign, but because there is far too little data to train it well.

> **What makes it stronger / weaker**
>
> **Stronger:** a proper temporal model that captures order and long-range dependence; the architecture that scales best as data grows. **Weaker:** data-hungry and high-variance at this scale; a long recurrence over a high-dimensional input is hard to pin down on 13 examples per class.

5 Role in the hybrid

The BiLSTM is included in the soft-vote ensemble but is **not** part of the recommended two-stream hybrid, which pairs the graph (ST-GCN) and attention (Transformer) streams instead. The reason is variance: the two-stream design deliberately combines the *lowest-variance* deep model (ST-GCN) with the *most expressive* one (Transformer). The BiLSTM's high variance would add noise rather than a complementary signal. It becomes a strong candidate again once the dataset is large enough to tame it.